

Digital product passports with intelligent metal sorting for end-of-life vehicle recycling.

Manufacturing × Product traceability and digital product passport × Machine learning · OS030 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
62 is the world moving	74 can we play, can we win	MEDIUM 0 named in the evidence	€309m observed evidence · per year	NEXT regulation adopted or propose...	L1 13 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity targets automotive manufacturers and recyclers facing the EU's Ecodesign for Sustainable Products Regulation (ESPR), which mandates Digital Product Passports for electronics and appliances by 2027. By combining DPPs with intelligent metal sorting, Orange can help customers automate compliance and improve recycling efficiency for end-of-life vehicles. The time is now because European technical standards for DPPs are in place, making implementation feasible.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€431m €77.6m – €2.3bn	€309m €55.7m – €1.7bn	€14.9m €2.7m – €79.8m	observed
Observed: tendered contract value, annualised	€20.9m €20.9m – €56.6m	€20.9m €20.9m – €56.6m	€1.0m €1.0m – €2.7m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (EU27_2020)	238,122 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Machine learning	3.3 % of enterprises (10+ employees)	observed	Eurostat · isoc_eb_ain2 · E_AI_TML	2025
Annual value of one engagement	€337k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	39,220 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	4.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

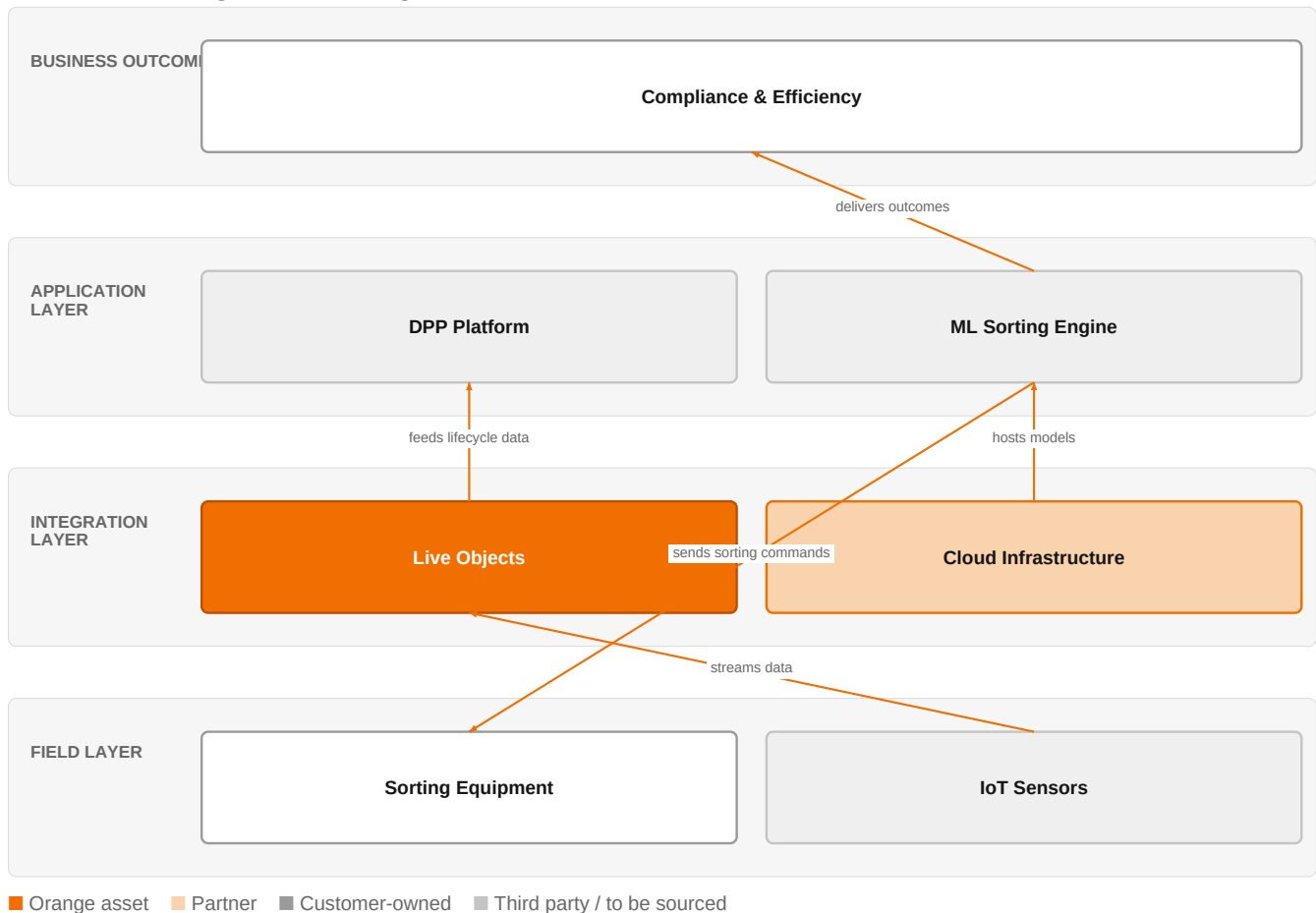
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€20.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	4.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

DPP-Enabled Intelligent Metal Sorting



This diagram shows how Orange integrates IoT, DPP, and ML to automate compliance and optimize metal sorting for end-of-life vehicles.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The EU's Ecodesign for Sustainable Products Regulation (ESPR) mandates Digital Product Passports for electronics and appliances by 2027, creating an urgent need for machine-readable lifecycle data. European technical standards for DPPs are now in place, and a DPP registry is live, signaling a shift from concept to operational reality. Current ELV recycling suffers from information gaps and manual compliance validation, which DPPs can address by enabling part-level traceability and automated compliance.

Sources: SIG-BA196FF05244 Springer Science and B 2026-02-03; SIG-AF45FABC720F RECYCLING magazin 2026-08-03; SIG-22FBEEA6354A Recyclingportal 2026-08-03; SIG-4FE21FD2ED06 openalex.org 2026-01-01

Who buys, and why now

The buyer is typically the head of sustainability or the chief operations officer at an automotive manufacturer or a large recycling operator. The pain is felt by recycling plant managers who struggle with inefficient sorting due to lack of detailed material data, and by compliance officers who face manual, error-prone validation. The trigger is the approaching ESPR compliance deadline, which forces them to implement DPPs or risk regulatory penalties.

Why it is hot now — every claim bound to a dated source

- Digital Product Passports can improve management and recycling of complex metal-rich products like end-of-life vehicles. [SIG-4FE21FD2ED06]
- DPPs are used to improve management and recycling of complex metal-rich products like end-of-life vehicles, addressing part-level traceability and automated compliance. [SIG-4FE21FD2ED06]
- Digital Product Passports improve management and recycling of complex metal-rich products like end-of-life vehicles. [SIG-4FE21FD2ED06]

- Current ELV recycling suffers from information gaps and manual compliance validation, which DPPs address.

[SIG-4FE21FD2ED06]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its IoT experts and Live Objects platform to collect and manage product lifecycle data from vehicles. AI/Data/Cloud experts would build machine learning models for intelligent metal sorting, leveraging Orange's cloud partnerships with AWS, Google Cloud, and NVIDIA for scalable infrastructure. The engagement would start with a pilot on a specific vehicle model, integrating DPP data with sorting equipment, then scale across the customer's fleet.

Why Orange can win

Orange can win because it combines IoT connectivity expertise with AI/Data/Cloud capabilities and holds ISO 27001 and TISAX certifications, which are critical for automotive data security. Its partnerships with hyperscalers like AWS and Google Cloud allow it to offer a flexible, secure cloud foundation. However, the assets are thin in terms of specific DPP or recycling domain expertise, so Orange must rely on its ability to integrate and orchestrate.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	L1	offer provides the technology in a matching domain	unconfirmed
AWS	partner	L2	partner provides the required technology	unconfirmed
Google Cloud	partner	L2	partner provides the required technology	unconfirmed
NICE	partner	L2	partner provides the required technology	unconfirmed
NVIDIA	partner	L2	partner provides the required technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed
Saint-Gobain Glass	reference	SUP	published reference in this vertical for this use case	unconfirmed
Skytale	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

MEDIUM competitive intensity (score 7.1; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	sells Machine learning; competes in OX: Smart Industries — also an Orange partner	structural	—
Google Cloud	Hyperscaler	sells Machine learning — also an Orange partner	structural	—
Atos / Eviden	Global systems integrator	sells Machine learning	structural	—

NICE	Customer-experience platform	sells Machine learning — also an Orange partner	structural	—
Databricks	Data and AI platform	sells Machine learning	structural	—
Dataiku	Data and AI platform	sells Machine learning	structural	—
Palantir	Data and AI platform	sells Machine learning; competes in OX: Smart Industries	structural	—
Accenture	Global systems integrator	active in Manufacturing; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 How are you currently managing end-of-life vehicle data for recycling compliance, and what gaps exist in part-level traceability?
- 2 Have you started planning for Digital Product Passport implementation under ESPR, and if so, what is your timeline?
- 3 What is your current metal sorting process, and how much manual intervention is involved in identifying and separating materials?
- 4 Do you have existing IoT or sensor infrastructure on your recycling lines that could feed data into a DPP system?
- 5 Which internal team would own the DPP initiative—sustainability, IT, or operations—and who has budget authority?
- 6 Are you already working with any cloud providers for data analytics, and what are your data sovereignty requirements?

Objections you will meet

They will say	You can answer
We already have a DPP solution from a specialist vendor.	That's a valid point, and many specialists offer strong DPP capabilities. However, our value lies in integrating that DPP with your existing IoT infrastructure and applying machine learning to optimize sorting—something a specialist may not cover. We can complement rather than replace.
The ESPR mandate doesn't apply to vehicles yet, so why invest now?	You're right that the current mandate covers electronics and appliances, but the regulation is expected to expand, and early movers will have a competitive advantage. Also, the operational benefits—reduced sorting costs and improved compliance—are immediate, regardless of regulatory timing.
We're concerned about data security and sovereignty.	That's a legitimate concern. Orange holds ISO 27001 and TISAX certifications, and we can deploy on sovereign cloud infrastructure through our partnerships with AWS and Google Cloud, ensuring your data meets automotive industry standards.

Risks and unknowns

The main risk is regulatory uncertainty: while ESPR mandates DPPs for electronics and appliances by 2027, it is not yet clear if and when end-of-life vehicles will be included. Additionally, the technical standards for DPPs are new, and interoperability across the supply chain is still evolving. There is also a risk that customers may prefer to work with a specialist DPP provider rather than a generalist integrator.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a European automotive OEM or recycler, differentiating Orange's Live Objects platform with ISO 27001 and TISAX certifications, and referencing our work with Saint-Gobain Glass on traceability and sustainability.
Sales	In a meeting with a European automotive manufacturer's sustainability or circular-economy lead, say: 'Our Live Objects IoT platform, combined with our AI and data expertise, can help you meet upcoming digital product passport requirements for end-of-life vehicles, improving recycling efficiency and compliance.'
Strategist / Innovator	Commission a deep-dive brief on how Live Objects plus AI/Data/Cloud experts could underpin a digital product passport for end-of-life vehicles, focusing on part-level traceability and automated compliance validation.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-36FFB2435F8A	2026-04-20	openalex.org	1	A Comprehensive Review of Digital Product Passports in Sustainable Manufacturing: Current State...
SIG-F47C31AB60CB	2026-02-14	MDPI AG	1	Dynamic Traceability and Reliability Management: Integrating the Digital Product Passport with ...
SIG-BA196FF05244	2026-02-03	Springer Science and Busi...	1	Digital Product Passport for Circular Economy: A Review of Semantic Web Vocabularies and Requir...
SIG-7B174A2FD055	2026-01-29	MDPI AG	1	Verisav Vocabularies: RDF/OWL Ontologies for Digital Product Passport and After-Sales Service A...
SIG-4FE21FD2ED06	2026-01-01	openalex.org	1	On the Utilization of Digital Product Passports for Intelligent Metal Sorting and Recycling
SIG-CC28A63D0BB7	2026-01-01	openalex.org	1	From Compliance to Circularity: Stakeholder Perspectives on Adopting Digital Product Passports ...
SIG-DA2C77C45DF6	2026-01-01	Fashion Works	1	The Digital Product Passport
SIG-8123897AE6E6	2025-12-15	openalex.org	1	Bridging digital tools and human skills: A sustainable competence framework for implementing di...
SIG-2B47C056B7E5	2025-10-30	openalex.org	1	Digital material passports in the AEC industry: a scoping review and conceptualisation of futur...
SIG-22FBEEA6354A	2026-08-03	Recyclingportal	2	The Digital Product Passport Registry is now live - Recyclingportal
SIG-AF45FABC720F	2026-08-03	RECYCLING magazin	2	Digital Product Passport: European Technical Standards Now in Place - RECYCLING magazin
SIG-26683D93E6D5	2026-05-25	HUB Style	2	Digital Product Passport 2027: nuova era per moda e luxury resale - HUB Style
SIG-4FE0689B82DF	2026-04-30	Packaging Digest	2	EU Digital Product Passport Impact on US Supply Chains - Packaging Digest

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:42:12+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.